

Tutorial 2/Module 1

DC Circuit Analysis and Theorems

30/07/2026

Dr. Sudipta Kumar Sarkar

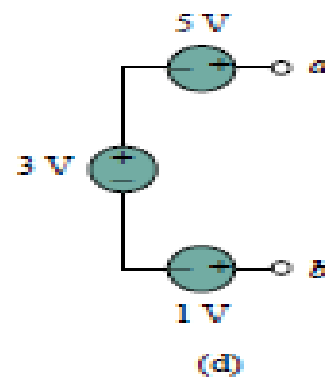
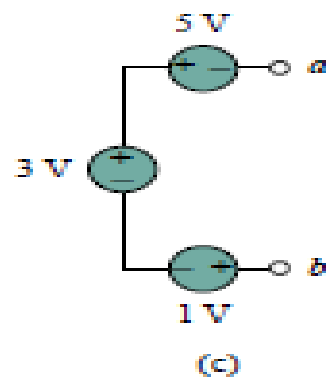
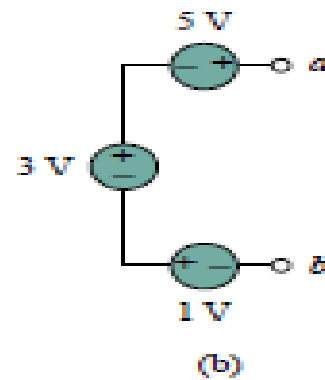
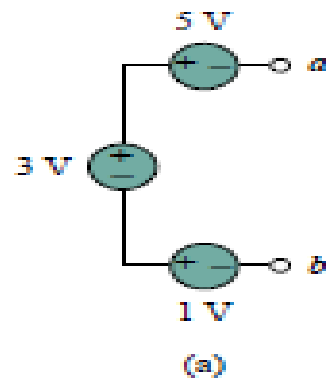
Assistant Professor (Research), SENSE



School of Electronics Engineering (SENSE)
VIT, Vellore, India

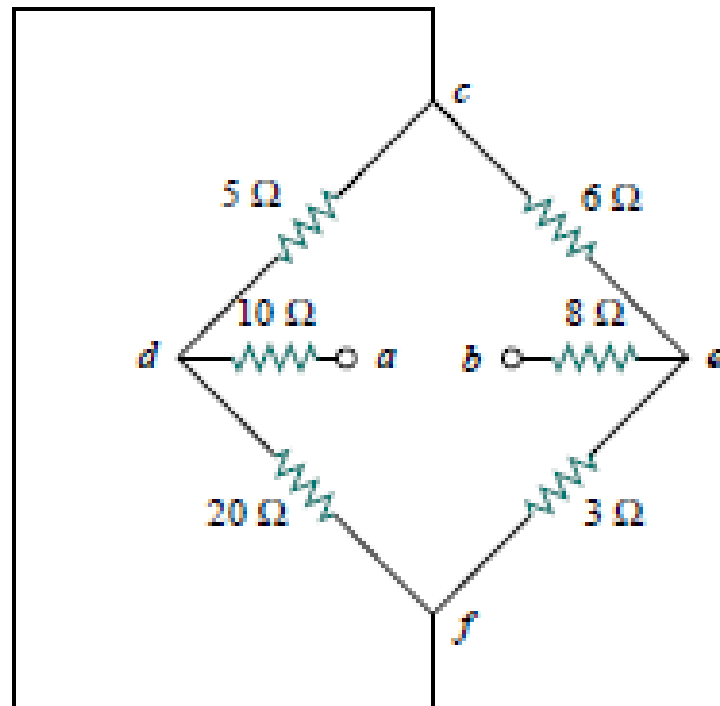
Problem 1

Which of the circuits in Fig. 2.66 will give you $V_{ab} = 7\text{ V}$?



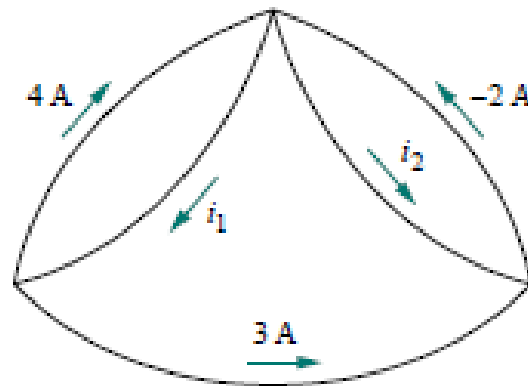
Problem 5

Find the equivalent resistance R_{ab} in the circuit of Fig. 2.106.



Problem 2

Determine i_1 and i_2 in the circuit in Fig. 2.74.



Problem 3

Find V_o in the circuit in Fig. 2.83 and the power dissipated by the controlled source.

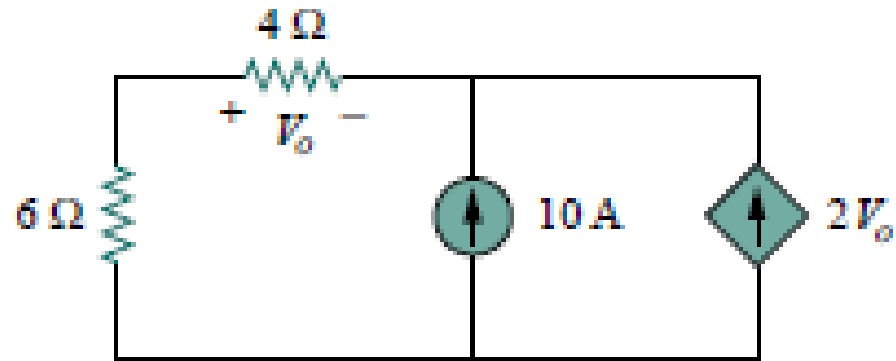


Fig. 2.83

70W

Problem 4

- 2.34 Find I and V_s in the circuit of Fig. 2.98 if the current through the $3\text{-}\Omega$ resistor is 2 A .

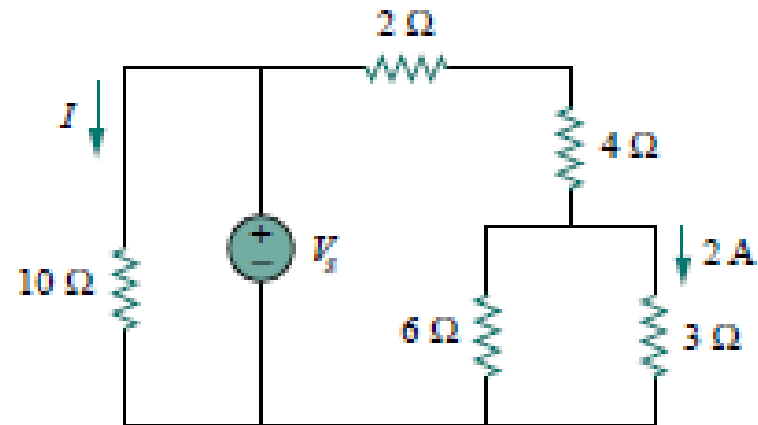
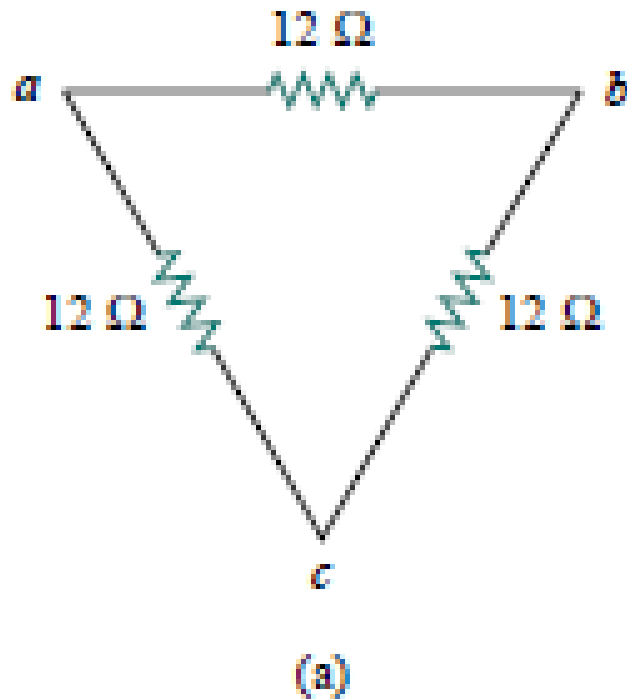


Figure 2.98 For Prob. 2.34.

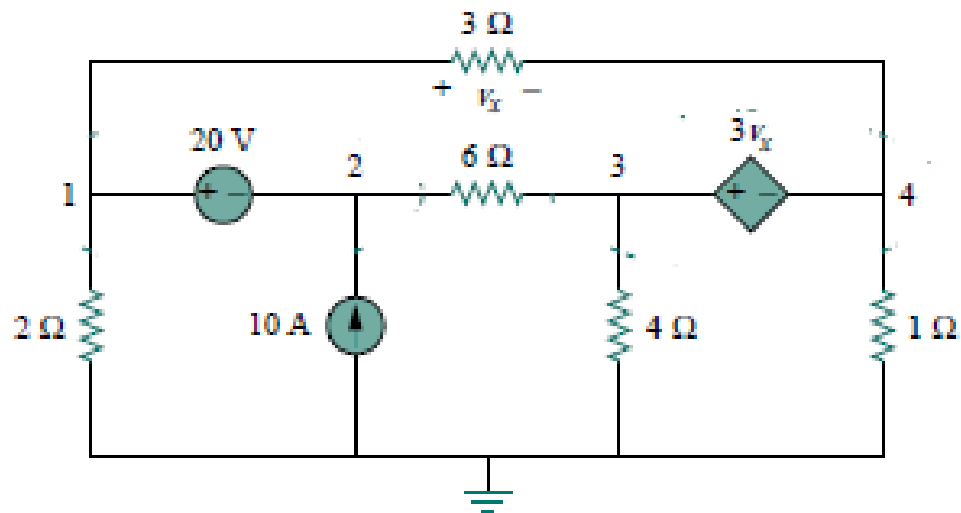
Problem 6



Y (??)

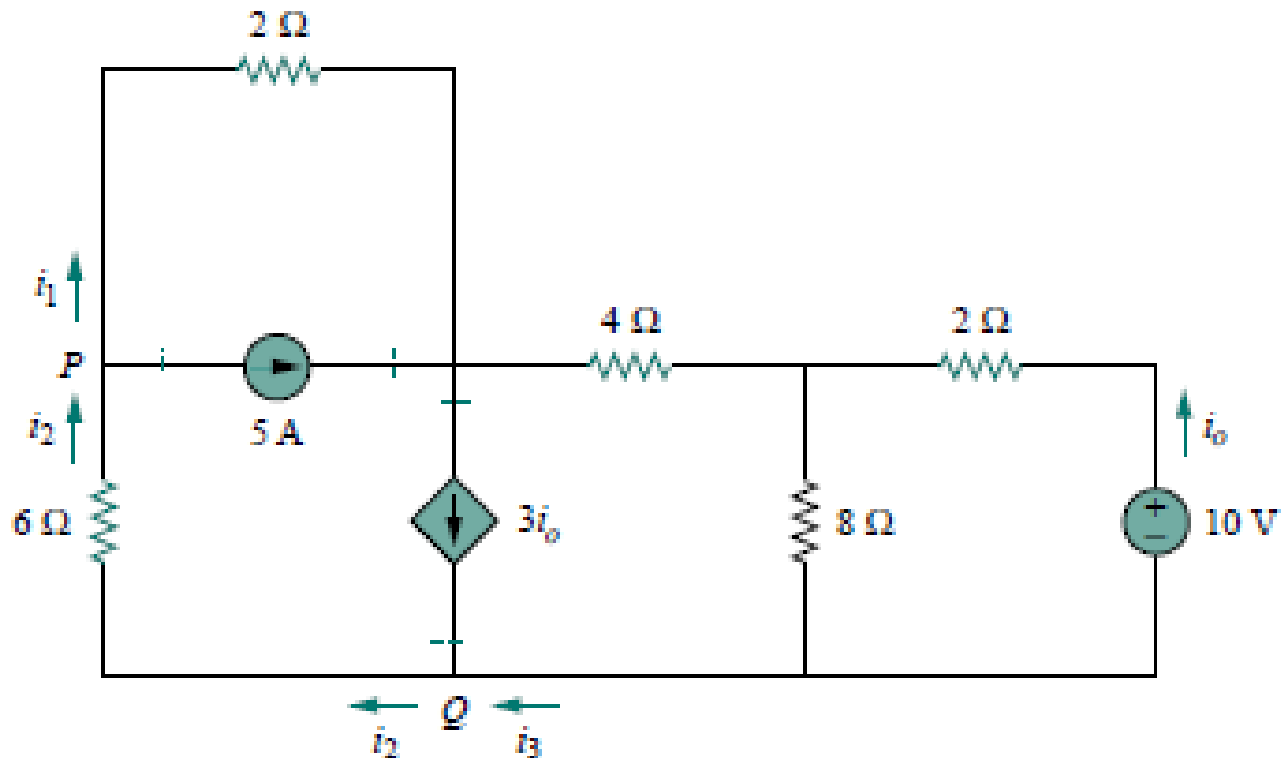
Problem 7

Find the node voltages in the circuit of Fig. 3.12.



Problem 8

For the circuit in Fig. 3.24, find i_1 to i_4 using mesh analysis.



Thank You